

# Reflection and Traceability Report on Software Engineering

Team #12, Streamliners

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[Reflection is an important component of getting the full benefits from a learning experience. Besides the intrinsic benefits of reflection, this document will be used to help the TAs grade how well your team responded to feedback. Therefore, traceability between Revision 0 and Revision 1 is an important part of the reflection exercise. In addition, several CEAB (Canadian Engineering Accreditation Board) Learning Outcomes (LOs) will be assessed based on your reflections. —TPLT]

## 1 Changes in Response to Feedback

[Summarize the changes made over the course of the project in response to feedback from TAs, the instructor, teammates, other teams, the project supervisor (if present), and from user testers. —TPLT]

[For those teams with an external supervisor, please highlight how the feedback from the supervisor shaped your project. In particular, you should highlight the supervisor's response to your Rev 0 demonstration to them. —TPLT]

[Version control can make the summary relatively easy, if you used issues and meaningful commits. If your feedback is in an issue, and you responded in the issue tracker, you can point to the issue as part of explaining your changes. If addressing the issue required changes to code or documentation, you can point to the specific commit that made the changes. Although the links are helpful for the details, you should include a label for each item of feedback so that the reader has an idea of what each item is about without the need to click on everything to find out. —TPLT]

[If you were not organized with your commits, traceability between feedback and commits will not be feasible to capture after the fact. You will instead need to spend time writing down a summary of the changes made in response to each item of feedback. —TPLT]

[You should address EVERY item of feedback. A table or itemized list is recommended. You should record every item of feedback, along with the source of that feedback and the change you made in response to that feedback. The response can be a change to your documentation, code, or development process. The response can also be the reason why no changes were made in response to the feedback. To make this information manageable, you will record the feedback and response separately for each deliverable in the sections that follow. —TPLT]

[If the feedback is general or incomplete, the TA (or instructor) will not be able to grade your response to feedback. In that case your grade on this document, and likely the Revision 1 versions of the other documents will be low. —TPLT]

## 1.1 Problem Statement and Goals

Table 1: Problem Statement and Goals — Feedback Response Summary

| Source    | Feedback Item (Brief)                                                           | Change Made (Brief)                                                                                              |
|-----------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| TA Rubric | Main objective/focus unclear; integrated project context missing                | Added explicit project objective, scope focus, and integration context in problem statement.                     |
| TA Rubric | Missing discussion: security/privacy, platform scope, integrations, performance | Added security/privacy concerns, mobile+web scope, third-party integration notes, and high-traffic expectations. |
| TA Rubric | Stakeholder hierarchy too rigid/over-structured                                 | Replaced primary/secondary/tertiary hierarchy with a single stakeholder list reflecting overlaps.                |
| TA Rubric | Goals not clearly measurable; missing security/privacy goal                     | Rewrote goals with intent + measurable criteria + rationale; added Security & Privacy goal.                      |
| TA Rubric | Reflection should be individual (Q1/Q2), team-wide (Q3)                         | Restructured reflection appendix: Q1/Q2 per member; Q3 team response.                                            |

Table 2: Development Plan — Feedback Response Summary

| Source           | Feedback Item (Brief)                                        | Change Made (Brief)                                                                                        |
|------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| TA Rubric (LO13) | Section 10 tech choices lacked rationale/pros/cons           | Expanded “Expected Technology” with project-specific rationale, trade-offs, and limitations per tool/tech. |
| TA Rubric        | Roles missing technical responsibilities and overlap plan    | Added technical ownership (primary/secondary) across frontend, mobile, backend, DB, CI/CD.                 |
| TA Rubric        | Workflow not at Contributor’s-Guide level                    | Added PR/review expectations, labels, quality gates, and Definition of Done.                               |
| TA Rubric        | Scheduling section shallow; requested timetable-style detail | Added a high-level timetable table with phases and planned outcomes.                                       |
| TA Rubric        | POC plan needed more context + failed-POC consequences       | Added project context, expanded risks, added scenario-based demo + failure-mode check, defined failed POC. |
| TA Rubric        | Formatting: justified alignment not applied in places        | Removed <code>flushleft</code> usage in communication plan section to avoid forced left alignment.         |
| TA Rubric        | GitHub link should point to main repo page                   | Updated link from board to repository URL.                                                                 |
| TA Rubric        | Reflection should be individual (Q1/Q2), team-wide (Q3)      | Restructured reflection appendix: Q1/Q2 per member; Q3 team response.                                      |

## 1.2 Development Plan and Proof of Concept Demonstration Plan

## 1.3 SRS and Hazard Analysis

## 1.4 Design and Design Documentation

### Module Interface Specification (MIS).

In response to design-review feedback, we strengthened the MIS so it better supports an independent implementer and stands alone as a document: the symbol and acronym table was expanded and explicitly scoped to this deliverable (no deferring readers to the SRS or MG for definitions). We added a short subsection stating how each module section is intended to be sufficient to build from (interfaces, typed state, assumptions, behaviour). Payment (M1), sign-up (M3), and access control (M5) state was rewritten using explicit data structures—for example consolidated payment and refund records, a named `SignUpRecords` map, and versioned policy storage (`policyByVersion`, `currentPolicyVersion`) with correct `roleAssignments` semantics. Other edits included tightening the module hierarchy wording, stating session-timeout expectations inside the MIS instead of pointing to other docs, replacing appendix placeholder text, and documenting the History Manager work for Team 10 with a public pull request: <https://github.com/OmarHassanAdelhamid/Five-of-a-Kind-capstone-project-/pull/328>. Unused SRS cross-document label imports were removed from the MIS source.

### Module Guide (MG).

We added brief narrative before each user-interface figure to explain what the screenshot shows and which modules it relates to, fixed a caption typo, and noted that abbreviations in the MG are defined in that document itself for consistency with the MIS approach.

## 1.5 VnV Plan and Report

### VnV Report:

- **Traceability readability feedback:** FR and module traceability matrices were hard to read when scaled. **Change made:** moved matrices to become larger and wide matrices to landscape pages instead of shrinking width\resizebox, improving readability without reducing font size.
- **NFR consistency feedback:** report NFR labels did not consistently match the VnV Plan. **Change made:** aligned the report to plan IDs NFR1–NFR6 and added a dedicated NFR trace crosswalk table for one-to-one mapping.
- **Scope/rigour feedback:** earlier wording implied full equivalence to planned load/usability activities. **Change made:** explicitly documented where lighter methods were used, what was partially verified, and what

remains as scope gaps (for example webhook idempotency, reliability monitoring, and audit depth).

- **Rev 0 product feedback:** “request refund not yet working” and “admin cannot edit created event.” **Change made:** implemented backend refund endpoint and frontend refund workflow, and replaced admin event-edit placeholder with a working edit form.

#### **VnV Plan:**

In response to TA feedback, we expanded the symbols and acronyms table (beyond the placeholder row), removed instructor template prompts (**wss**) and replaced them with final text, and tightened vague objectives and usability language—including concrete definitions for elevated load (tied to existing NFR concurrency cases) and measurable usability criteria (task success, timings, Likert item for NFR3). The appendix now documents symbolic parameters from those tests and states how optional usability surveys relate to NFR3. After the Module Interface Specification (MIS) was available, we filled the unit-test section: implementation verification references modules M1–M8; for each module we added representative unit test cases (**UT-M\***) mapped to MIS access programs and exceptions, plus a traceability table to SRS areas. Revision history records version 1.1 (MIS-dependent content) and 1.2 (TA feedback pass).

## **2 Extras**

[Summarize the extras that were tackled by this project. Extras can include usability testing, code walkthroughs, user documentation, formal proof, Gender-Mag personas, Design Thinking, etc. Extras should have already been approved by the course instructor as included in your problem statement. —TPLT]

## **3 Design Iteration (LO11 (PrototypeIterate))**

[Explain how you arrived at your final design and implementation. How did the design evolve from the first version to the final version? —TPLT]

[Don’t just say what you changed, say why you changed it. The needs of the client should be part of the explanation. For example, if you made changes in response to usability testing, explain what the testing found and what changes it led to. —TPLT]

## **4 Design Decisions (LO12)**

[Reflect and justify your design decisions. How did limitations, assumptions, and constraints influence your decisions? Discuss each of these separately. —TPLT]

## 5 Economic Considerations (LO23)

[Is there a market for your product? What would be involved in marketing your product? What is your estimate of the cost to produce a version that you could sell? What would you charge for your product? How many units would you have to sell to make money? If your product isn't something that would be sold, like an open source project, how would you go about attracting users? How many potential users currently exist? —TPLT]

## 6 Reflection on Project Management (LO24)

[This question focuses on processes and tools used for project management. —TPLT]

### 6.1 How Does Your Project Management Compare to Your Development Plan

[Did you follow your Development plan, with respect to the team meeting plan, team communication plan, team member roles and workflow plan. Did you use the technology you planned on using? —TPLT]

### 6.2 What Went Well?

[What went well for your project management in terms of processes and technology? —TPLT]

### 6.3 What Went Wrong?

[What went wrong in terms of processes and technology? —TPLT]

### 6.4 What Would you Do Differently Next Time?

[What will you do differently for your next project? —TPLT]

## 7 Ethics and Equity (LO18)

[Describe how your team applied the principle of equity and universal design to ensure equitable treatment of all stakeholders. —TPLT]

## 8 Reflection on Capstone

[This question focuses on what you learned during the course of the capstone project. —TPLT]

## **8.1 Which Courses Were Relevant**

[Which of the courses you have taken were relevant for the capstone project?  
—TPLT]

## **8.2 Knowledge/Skills Outside of Courses**

[What skills/knowledge did you need to acquire for your capstone project that  
was outside of the courses you took? —TPLT]